OS Orange Problem

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PES1UG24AM087

1A)

```
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

1B)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

2A)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

2B)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ xxd .pes/objects/4d/51edae2d36944dea5fcf238a5624f53f
951afd0aeeae0f1ac38c7b29f0b559 | head -20
00000000: 636f 6d6d 6974 2031 3930 0074 7265 6520  commit 190.tree
00000010: 6264 3962 3736 3962 6234 6439 3931 3063  bd9b769bb4d9910c
00000020: 3331 6234 6630 6539 3965 3433 3735 6666  31b4f0e99e4375ff
00000030: 6665 6133 3563 3230 3535 3133 3063 3234  fea35c2055130c24
00000040: 3866 6631 3331 6637 3863 3632 3634 3432  8ff131f78c626442
00000050: 0a61 7574 686f 7220 4468 7275 7620 3c64  .author Dhruv <d
00000060: 6872 7576 7461 6c61 7661 7433 3640 676d  hruvtalavat36@gm
00000070: 6169 6c2e 636f 6d3e 2031 3737 3638 3333  ail.com> 1776833
00000080: 3634 380a 636f 6d6d 6974 7465 7220 4468  648.committer Dh
00000090: 7275 7620 3c64 6872 7576 7461 6c61 7661  ruv <dhruvtalava
000000a0: 7433 3640 676d 6169 6c2e 636f 6d3e 2031  t36@gmail.com> 1
000000b0: 3737 3638 3333 3634 380a 0a49 6e69 7469  776833648..Initi
000000c0: 616c 2063 6f6d 6d69 74                   al commit
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

3A)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt && echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     bye.txt
  staged:     file1.txt
  staged:     file2.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test.txt
  untracked:  commit.c
  untracked:  pes.c
  untracked:  commit.h
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  tree.h
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  .gitignore
  untracked:  test_objects.c
  untracked:  tree.c
  untracked:  index.c
```

3B)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ cat .pes/index
100644 e67ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc 1776833673 8 bye.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776834003 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776834003 6 file2.txt
100644 05c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43 1776833673 12 hello.txt
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

4A)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Committed: 7eb405ffab7e... Add world
Committed: e3d23544c127... Add farewell
commit e3d23544c127bec641993345c426d7be123a46f210f64a8883e2fb04a61f34fd
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833673

    Add farewellauth!

commit 7eb405ffab7ee9fef32de25eb13bffa4d5569c9a2a60929c074d9fc192d6b911
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833673

    Add worldellauth!

commit 4d51edae2d36944dea5fcf238a5624f53f951afd0aeeae0f1ac38c7b29f0b559
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833648

    Initial committial commiA
```

4B)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/4d/51edae2d36944dea5fcf238a5624f53f951afd0aeeae0f1ac38c7b29f0b559
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7e/b405ffab7ee9fef32de25eb13bffa4d5569c9a2a60929c074d9fc192d6b911
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e3/d23544c127bec641993345c426d7be123a46f210f64a8883e2fb04a61f34fd
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

4C)

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
e3d23544c127bec641993345c426d7be123a46f210f64a8883e2fb04a61f34fd
ref: refs/heads/main
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

Final Screenshot

```
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 066289d9e5ca... Initial commit

Log after first commit:
commit 066289d9e5cac6e8ef1951f47a7e0b2a4728bcd4905033099ea5d93977aa9cde
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833774

    Initial commit


--- Second Commit ---
Committed: dc6547b2849f... Update file.txt

--- Third Commit ---
Committed: 6845ae8813de... Add farewell

--- Full History ---
commit 6845ae8813de6e1e9d75bb70769972aed6d943be9a8d1cc56e7ecc0365186c15
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833774

    Add farewellauth!

commit dc6547b2849ffbb77803d5ec82c069735cfeacf4a5339458feae60ed495de575
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833774

    Update file.txth!

commit 066289d9e5cac6e8ef1951f47a7e0b2a4728bcd4905033099ea5d93977aa9cde
Author: Dhruv <dhruvtalavat36@gmail.com>
Date:   1776833774

    Initial committial commiA


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6845ae8813de6e1e9d75bb70769972aed6d943be9a8d1cc56e7ecc0365186c15

--- Object Store ---
Objects created:
10
.pes/objects/06/6289d9e5cac6e8ef1951f47a7e0b2a4728bcd4905033099ea5d93977aa9cde
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/68/45ae8813de6e1e9d75bb70769972aed6d943be9a8d1cc56e7ecc0365186c15
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/dc/6547b2849ffbb77803d5ec82c069735cfeacf4a5339458feae60ed495de575

=== All integration tests completed ===
dhruv@Dhruv:~/PES1UG24AM087-pes-vcs$
```

Phase 5 & 6 — Analysis Questions (Written)

Q5.1 — checkout: You'd change .pes/HEAD to point to the new branch name, read the target branch's tree from its commit, then walk the tree recursively and overwrite every tracked file in the working directory. It's complex because you must handle files that exist in one branch but not the other (delete them) and detect conflicts.

Q5.2 — Dirty working directory: For each file tracked in the index, stat it and compare mtime/size to the index entry. If they differ, recompute the blob hash and compare to the index hash. If there's a mismatch AND the index entry differs between the two branches' trees, there's a conflict → abort checkout.

Q5.3 — Detached HEAD: New commits are written but no branch ref is updated. If you switch away, those commits become unreachable. To recover, git reflog shows all previous HEAD positions — you can create a branch at the orphaned commit hash to save them.

Q6.1 — GC: Walk all branches and tags as roots, do a DFS/BFS following commit → parent, commit → tree, tree → subtrees/blobs. Use a HashSet of all reachable object IDs. Then list all files in .pes/objects/ and delete any not in the set. For 100k commits × 50 branches you'd visit roughly 100k–500k objects (commits + trees + blobs), bounded by deduplication.

Q6.2 — GC race condition: A commit operation may: (1) write a new blob, (2) write a new tree, (3) write a new commit, (4) update the branch ref. If GC runs between steps 2 and 4, it sees the new tree/blob as unreachable (branch still points to old commit) and deletes them. The in-progress commit then completes with a dangling reference. Git avoids this using "loose object grace periods" — objects newer than a threshold (default 2 weeks) are never collected regardless of reachability.